Conceptual Topography

A Field-Theoretic Account of Conceptual Landscapes

VED × IFGT Application Layer 1

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This paper uses the VED×IFGT framework described in Endoh [2026b,c,d,e,a] as a source of formal vocabulary. Its notation was attributed to the Master Variable Table v2.1; the exact artifact and version behind that attribution remain unresolved. The present work does not redefine C, I, Φ, J_I, J_C, σ, or τ; it proposes cognitive uses of this vocabulary without claiming a typed derivation from the upstream theories. The primary definition $\Phi = K * I$ is used throughout; the linear approximation $\Phi \approx -\alpha C + \Phi_0$ is employed only where explicitly stated. Semantic, psychological, or intentional interpretations of these quantities are not assumed unless explicitly stated. The strict inequality $C < C_{\max}$ is retained as a model constraint. Conceptual-domain uses of I, C, and Φ are proposed projected or coarse-grained interpretations within a cognitive observation window. Shared notation does not establish shared quantities or a typed translation. Formal conditions, including the scope of the force relation, positivity preservation, token-coupling signatures, and the long-time limit, remain unresolved where not specified.

The definitions below fix local model vocabulary. Semantic and cognitive readings are interpretations; phenomenal readings are proposed correspondences, not explanations of subjective experience. The paper supplies neither operational mappings nor model-linked empirical validation. The existing figures and their captions are scoped as conceptual schematics, not measured trajectories, executed dynamics, or evidence for the proposed interpretations.

Abstract

This paper presents Conceptual Topography (CT), a field-theoretic account of conceptual landscapes, as the first application layer of the VED×IFGT framework. Conceptual landscape is a model-local designation for the proposed long-time geometry of the information potential field $\Phi(x, \tau)$ under recursive token–field interaction. Sedimentation, its relation to thought, and the invisible-terrain picture are interpretations, not established convergence or phenomenal-access results.

Within the VED×IFGT framework, CT organizes a proposal around three coupled processes. First, the continuous information field $I(x, \tau)$ is used in a proposed cognitive-window interpretation of the VED×IFGT relation $I = f(1 - C)$, with effective potential $\Phi = K * I$ providing the geometry used to represent passability. Second, discrete language tokens $\ell_i = \mathcal{Q}[I, \Phi, C;$ constraints$]$ act as re-addressable discrete units assigned a local interaction with $\nabla\Phi$; well deepening through reuse is a qualitative mechanism hypothesis. Third, the recursive loop $\Delta \to C \to I \to \Phi \to J_I \to \Delta$ describes a proposed feedback architecture. The sedimentation expression $\Phi \xrightarrow{\text{sed.}} \Phi^*$ is retained as a long-time template, with its convergence and history-dependent interpretation unresolved.

The local structural definitions are retained: a concept is a locally stable region of Φ satisfying $\nabla\Phi(x) \approx 0$; understanding is a temporary reduction in traversal cost across such a region. Thought is a proposed cognitive reading of trajectories associated with the formal flow field J_I, not identical to that field. Within the model, strict positivity $I(x, \tau) > 0$ is interpreted as excluding ordinary complete conceptual closure. Further differentiation

is proposed under residual asymmetry, boundary structure, or new perturbations, not guaranteed by positivity alone.

At the social scale, CT compares token functions and dominant costs. Shared terrain, institutional attractors, and collective maintenance are proposed social interpretations; identical individual/collective dynamics and institutional explanations are not established.

This work does not redefine any variable of the VED×IFGT framework; it proposes cognitive and linguistic uses of that vocabulary without establishing typed inheritance, operational mappings, or empirical validation.

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1 Introduction

This paper operates within the VED×IFGT framework [Endoh, 2026b,a] and proposes model-local interpretations in the domain of conceptual structure and thought dynamics. The intended notation continuity with the Master Variable Table v2.1 is retained, while its exact source/version and typed inheritance remain unresolved. The framework developed here is called Conceptual Topography (CT); the phrase "conceptual terrain" refers to the same object when the intuitive metaphor is emphasized.

1.1 The Problem of Conceptual Structure

What is a concept? The standard answer places concepts in the semantic domain: a concept is a meaning-bearing unit, a mental representation, or a node in a propositional network. This answer, while useful for many purposes, leaves the *dynamical* question unanswered. How does a concept form, stabilize, and constrain subsequent thought? How does understanding arise, and why does it sometimes fail to transfer even when the relevant definitions are available?

The present paper approaches these questions from a different direction. Rather than asking what a concept *means*, we ask what structural conditions allow a region of the information field to persist, be re-entered at low cost, and systematically bias future dynamics. This reframing shifts the problem from semantics to geometry.

1.2 Position in the VED×IFGT Series

The VED×IFGT framework consists of two foundational layers. Vortical Enclosure Dynamics (VED) describes the generation of structure from difference, gradient, flow, and closure [Endoh, 2026b]. Information Field Geometry Theory (IFGT) describes the complementary informational structure that persists as quasi-closure in the region where C does not reach unity [Endoh, 2026a].

The present work is positioned as *Application Layer 1* of this framework: it proposes cognitive and social interpretations of IFGT's field-theoretic vocabulary. It does not introduce new foundational variables; the VED×IFGT equation motivates candidate structural classes for organizing cognitive phenomena, not an established existence or classification theorem.

The series architecture is:

Layer	Theory
L0 (generation)	VED
L0 (structure)	IFGT
L1 (application)	**Conceptual Topography [this paper]**
L1 (application)	Consciousness Theory [repository companion]
L1 (application)	Intelligence Theory [repository companions]

1.3 Main Claims

The paper retains local definitions and proposes the following interpretations. Definitions of concept and understanding do not establish the cognitive or phenomenal correspondences discussed alongside them:

1. A **concept** is a locally stable region of the information potential $\Phi(x, \tau)$, defined by $\nabla\Phi(x) \approx 0$ with stability under small perturbations.

2. **Understanding** is a temporary reduction in traversal cost across a conceptual region, not the acquisition of correct meaning.

3. **Thought** is interpreted using trajectories associated with the formal information flow J_I. This description includes bias from $\nabla\Phi$, not a complete path specified by the gradient alone, and does not require a directing subject within the model.

4. **Language tokens** are modeled as discrete re-addressable units that couple locally to $\nabla\Phi$ without carrying semantic content. Deepening through reuse is a hypothesis, with typing and update consequences still unresolved.

5. Within the model, strict positivity $I(x, \tau) > 0$ is interpreted as excluding ordinary complete conceptual closure. Further differentiation is a conditional proposal, compatible with local quiescence and not implied pointwise by positivity alone.

6. At the **social scale**, a comparison of token roles motivates proposed shared terrain, institutional attractor analogies, and a local collective-maintenance description. A common dynamics or general definition of intelligence is not established.

1.4 Structure of the Paper

Section 2 presents an intuitive picture of conceptual landscapes. Section 3 proposes correspondences from VED×IFGT vocabulary to conceptual phenomena. Section 4 presents the existing field and coupling expressions with their local definitions and unresolved formal scope. Section 5 organizes candidate structural classes. Section 6 develops qualitative hypotheses of sedimentation, forgetting, and conceptual change. Section 7 treats the scale inversion from individual to social landscapes. Section 8 describes proposed failure regimes and limits. Section 9 connects the formal structure to the intuitive recorded note vocabulary, while preserving unresolved artifact attribution. Section 10 discusses implications and open questions. Section 11 concludes.

2 Intuitive Picture

Before introducing the formal apparatus, we sketch the intuitive picture that motivates the technical development.

2.1 Thought as Movement Across Terrain

Consider the experience of working through a difficult problem. Some directions of thought feel natural and low-cost; others require effort and tend to be abandoned. Some ideas return repeatedly without deliberate intention; others, once visited, leave no trace. We propose to interpret the sense of understanding in terms of a change in the ease of moving through a region of thought, rather than as the acquisition of a new object. This phenomenological correspondence is separate from the model's threshold definition of understanding.

These experiential examples motivate a spatial metaphor: thought moves across a terrain whose shape is not fixed in advance but is formed by the history of prior movement. In this proposed picture, frequently traversed paths become smoother, while less visited regions become less accessible. The invisible terrain is a metaphor for experiencing ease or difficulty without inspecting the proposed geometry, not a claim of proven inaccessibility.

This is the intuitive core of Conceptual Topography. Figure 1 summarizes this geometric picture.

The terrain is the model-local reading of the information potential $\Phi(x, \tau)$ as history-shaped geometry associated with information flow J_I. Movement is a trajectory representation associated with J_I, with $-\nabla\Phi$ contributing a directional bias rather than specifying the entire path. Calling this movement thought is a further cognitive interpretation, not a field/path/content

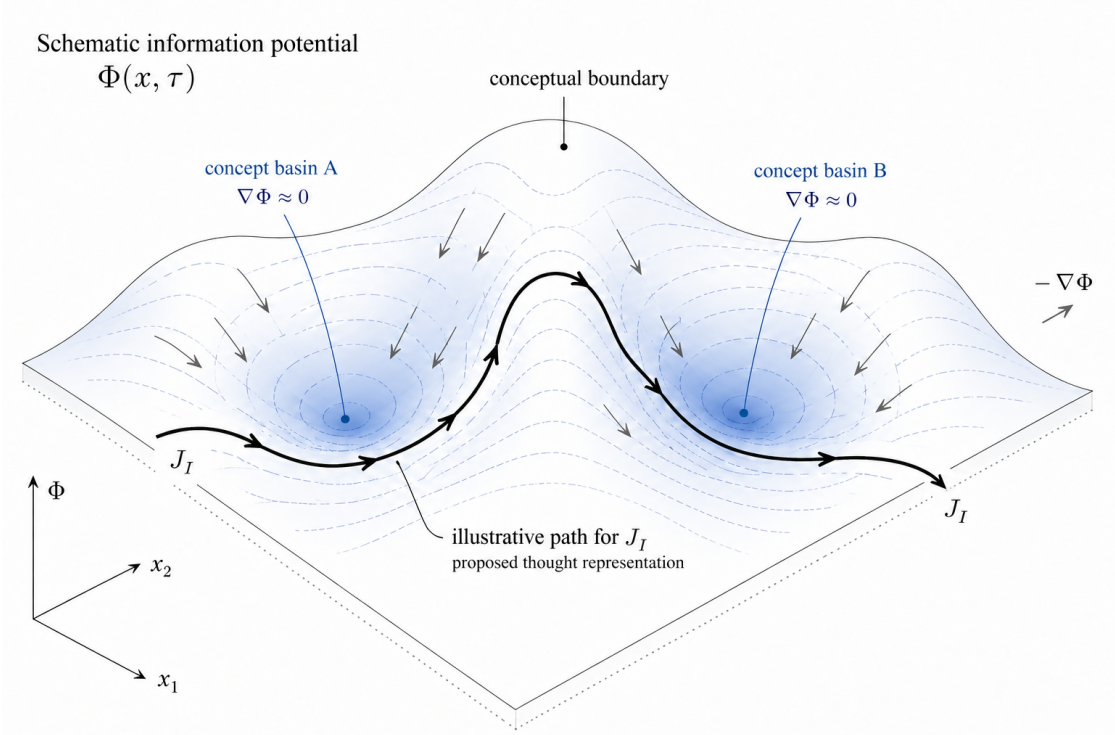


Figure 1: Schematic conceptual landscape as information-potential geometry. The surface illustrates the model potential $\Phi(x, \tau)$. Basins depict the local concept definition where $\nabla\Phi \approx 0$. The drawn path is an illustrative representation associated with the flow field J_I, not the field itself or a measured thought path. Its cognitive reading is proposed; ridges illustrate model-level conceptual boundaries.

identity. The understanding picture uses a local reduction in $|\nabla\Phi|$ in a previously difficult region. Its correspondence to felt ease remains a hypothesis: the formal threshold definition is not independent evidence for a sense of understanding.

2.2 Concepts as Valleys

In this picture, a concept is a valley in the potential landscape: a region of low Φ toward which trajectories are naturally drawn. Valley depth is proposed to reflect repeated traversal, and width is compared with the range of contexts in which a concept applies. These comparisons do not supply independent semantic measurements.

Crucially, the valley does not contain a meaning. It is described as a *dynamical attractor*: a region that biases subsequent trajectories in the model, with patterns of thought as the proposed cognitive interpretation. What we call the "meaning" of a concept is, in this framework, the set of trajectories that the attractor makes available.

2.3 Linguistic Tokens as Terrain-Shaping Operations

Linguistic tokens — understood broadly to include not only natural language words but any discrete, re-addressable medium capable of acting on the field as a localized perturbation: symbols, emblems, gestures, diagrams, mathematical notation, and so on — are not assigned intrinsic meanings in this account. They are modeled as localized perturbations that bias the landscape. Repeated use in similar contexts is hypothesized to deepen an associated valley, while declining reuse is associated with shallowing. More similar sender and receiver landscapes are proposed to support more similar responses to a perturbation; divergence may contribute

to communication failure. This comparative hypothesis neither exhausts the explanation of communication nor estimates geometry from communication success.

2.4 The Invisible GPU

The invisible-GPU metaphor proposes a structural interpretation of cognition without direct inspection of the landscape. It is used to organize experiential examples: thoughts coming naturally, connections suggesting themselves, and understanding arriving suddenly rather than being constructed step by step.

These experiences are proposed to correspond to the geometry of Φ, rather than to intrinsic meanings stored in that geometry. This is not an explanation of how subjective experience is generated.

The following sections distinguish this intuitive vocabulary from the formal definitions and the unresolved cognitive and phenomenal mappings.

3 Formal Mapping

This section records proposed correspondences between VED×IFGT variables and conceptual phenomena. The table separates formal roles from model-local interpretations; it does not supply operational mappings. The attribution to the Master Variable Table v2.1 remains subject to the source-provenance limitation stated at the beginning of the paper.

3.1 The Background Field

The information density field $I(x, \tau) = f(1 - C(x, \tau))$ is used as the continuous field in the model of conceptual structure. In this paper, I is not a new cognitive primitive but a proposed cognitive-window interpretation of the IFGT information-density vocabulary, not an established typed identity. In this local interpretation, I represents the density of unresolved informational tension at position x and evolution parameter τ: regions described as quasi-closed, neither fully stabilized nor fully dissipated.

High I is proposed to correspond to conceptually active regions; low I to regions where structure has largely closed. This is not a calibrated activity measure. Within this account, the strict positivity $I(x, \tau) > 0$ is interpreted as excluding complete closure as an ordinary conceptual state. It does not by itself guarantee a nonzero local gradient or flow. The role of residual asymmetry, nonuniform history, and boundary structure in gradient formation remains a model-level hypothesis.

The information potential

$$\Phi(x, \tau) = (K * I)(x, \tau) \tag{1}$$

is the primary geometric object. Its landscape reading is a model-level interpretation of accumulated bias. The potential gradient contributes an effective directional bias; it does not alone determine a complete trajectory, since the flow relation also contains a diffusion term. The field-to-trajectory-to-thought correspondence is not supplied as a typed or operational map.

3.2 Variable Correspondence Table

VED×IFGT variable	Conceptual domain interpretation	Formal role
$C(x,\tau)$	Degree of conceptual fixation at x	Background field; $C \in [0, C_{\max})$
$I(x,\tau) = f(1-C)$	Density of unresolved conceptual tension	Central field; $I > 0$ always
$\Phi = K * I$	Model-local conceptual landscape	Potential geometry
$\nabla\Phi$	Proposed bias toward lower cognitive resistance	Constraint force on J_I
$J_I = -D\nabla I + \mu F_I$	Proposed representation of thought	Formal flow field
$\sigma = \sigma_d - \sigma_p - \sigma_b$	Rate of landscape deformation	Source term
Δ	Novel experience, perception, or linguistic input	Driving source
δI	Unresolved conceptual residue	Trace that constitutes I
τ	Proposed experiential-time correspondence	Formal evolution / ordering parameter
$\ell_i = \mathcal{Q}[I, \Phi, C; \cdot]$	Reusable linguistic token (natural language word, symbol, emblem, gesture, diagram, formula, or any discrete unit capable of acting on the field as a localized perturbation)	Proposed action on Φ; signature unresolved
Φ^*	Mature, sedimented conceptual landscape	Long-time attractor of Φ
$C_{\max}$	Ideal complete-closure boundary	Boundary representation, not an ordinary state

The temporal parameter orders formal evolution; its experiential reading is a separate proposal, not a clock calibration. Likewise, the sedimented landscape and long-time-attractor labels retain the unresolved limit and history-dependent interpretation discussed in Section 4.

Remark 3.1 (Notational remark)**.** *The operator $\mathcal{Q}$ is called a quantization operator because its action — extracting discrete, re-addressable units from a continuous field configuration — is compared structurally with the passage from continuous variables to discrete values. This is an informal analogy, not a mathematical isomorphism or a shared physical operation. The term "quantum" is not used as a theoretical claim within CT, but the analogy is preserved in the name of this operator.*

3.3 What the Mapping Does Not Claim

The proposed correspondences above do not introduce semantic content, intentionality, or psychological states into the formalism. In particular:

- Φ is not a meaning field. It is a dynamical constraint geometry used to describe relative transition costs within the model.

- J_I is not thought content. It is the formal flow field; a trajectory description and its interpretation as thought are distinct uses. Content appearing in experience is treated as a downstream topic, not a consequence established by identifying these uses.
- ℓ_i is not a meaning unit. It is a discrete re-addressable unit that couples locally to $\nabla\Phi$.

These exclusions are methodological necessities, not claims about the ultimate nature of meaning or experience.

4 Core Equations

Formal scope and unresolved assumptions. The domain, geometry and measure, and parameter ranges are not fully specified. Additional assumptions and conventions in related IFGT formulations, including normalization choices, are not automatically adopted as CT specialization conditions. The scope statements below identify missing specifications; they do not supply them.

4.1 Field Evolution

The balance relation is used here as a model-level formal template. A complete initial-boundary-value problem, including the domain, initial and boundary conditions, regularity, and positivity-preservation conditions, is not specified. Use of the relation does not establish existence, uniqueness, or well-posedness:

$$\frac{\partial I}{\partial \tau} + \nabla \cdot J_I = \sigma, \tag{2}$$

where the information flow is

$$J_I = -D\nabla I + \mu F_I, \qquad F_I = -\nabla\Phi = \alpha \nabla C, \tag{3}$$

In the force relation, the first equality defines the effective force from the negative potential gradient. The subsequent closure-gradient equality is intended as a local/effective specialization, not as a derivation from the primary convolution, which remains the general field relation. The approximation regime and the assumptions under which that specialization is valid remain to be fixed.

The generation term decomposes as

$$\sigma = \sigma_{\text{drive}} - \sigma_{\text{dissip}} - \sigma_{\text{barrier}}. \tag{4}$$

The three contributions are:

$$\sigma_{\text{drive}} = a\Delta\left(1 - \frac{C}{C_*}\right), \tag{5}$$

$$\sigma_{\text{dissip}} = \Gamma C, \tag{6}$$

$$\sigma_{\text{barrier}} = \frac{\kappa_B}{C_{\max} - C}. \tag{7}$$

Here C_* denotes the effective saturation scale of conceptual quasi-closure within the observation window, while $C_{\max}$ denotes the ideal complete-closure boundary. In general $C_* < C_{\max}$ for conceptual regimes.

The following are effective cognitive interpretations, not measured mechanisms: σ_{drive} represents the influx of novel input (perception, linguistic perturbation, reward signal); σ_{dissip} represents the natural decay of unresolved tension (forgetting, habituation); σ_{barrier} represents a projected or coarse-grained effective resistance against conceptual overclosure. It should not be read as a fundamental external prohibition; it is the cognitive-domain analogue of the

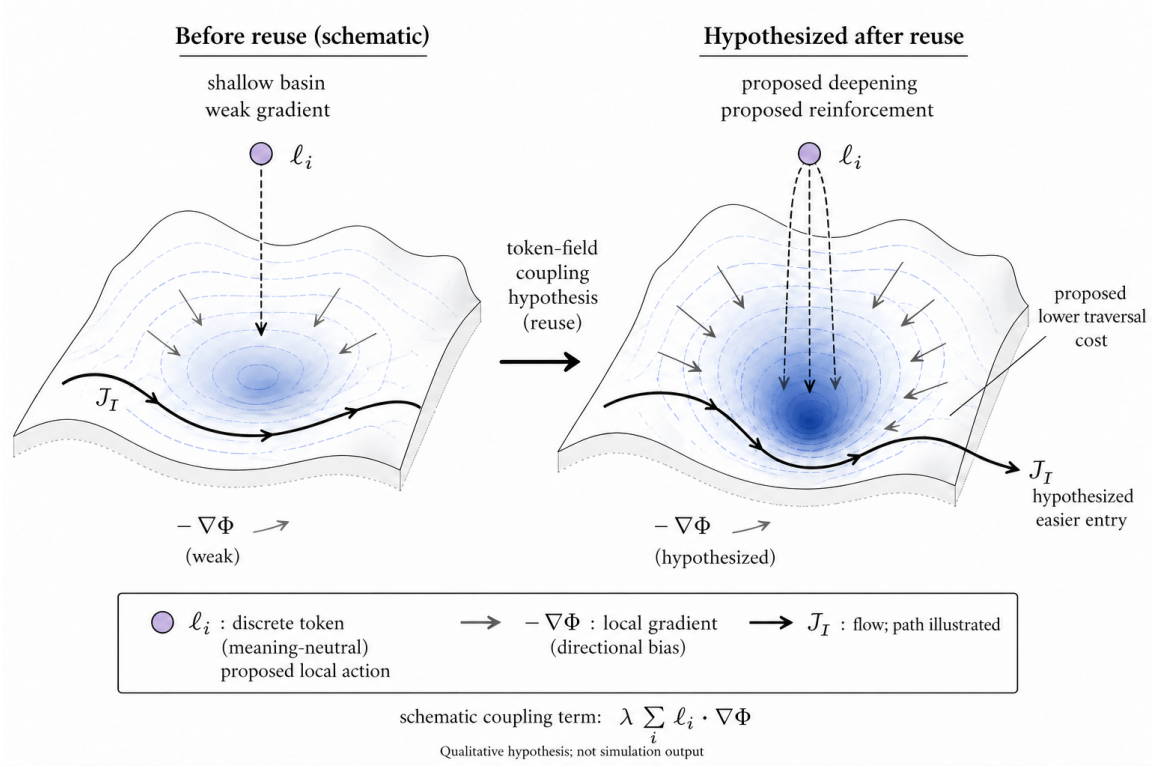


Figure 2: Schematic token–field coupling hypothesis, not simulation output. Discrete tokens ℓ_i are not assigned intrinsic meaning; their proposed local coupling to $\nabla\Phi$ motivates the illustration. Deepening and lower traversal cost for paths associated with J_I are hypothesized effects, not executed or derived results. The expression $\lambda \sum_i \ell_i \cdot \nabla\Phi$ remains schematic: token realization and pairing are unresolved.

effective barrier notation used in VED and IFGT. The reciprocal form does not by itself prove dynamical inaccessibility of the stated upper boundary. Divergence of a reciprocal expression and invariance under the full evolution are distinct claims; neither endpoint behavior nor invariant-domain conditions are established here.

4.2 Token–Field Coupling

The present architecture treats discrete, re-addressable structures as a proposed requirement for stable, low-cost recursion. The motivation is that revisiting a continuous field configuration may be costly and drift-prone; this is not a general impossibility claim about continuous systems. Re-addressability is intended to support reuse of complex states. Figure 2 illustrates the coupling hypothesis, not an executed demonstration of its consequences.

The medium that enables this structure is the linguistic field, where "linguistic" is understood broadly to include not only natural language but any discrete, re-addressable medium capable of acting on the field as a localized perturbation: symbols, emblems, gestures, diagrams, mathematical notation, and so on.

Linguistic tokens ℓ_i (discrete units drawn from any such medium) are assigned the following effective token–field coupling expression:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{field}}[I] + \mathcal{L}_{\text{token}}[\ell] + \lambda \sum_i \ell_i \cdot \nabla\Phi. \tag{8}$$

The proposed coupling role is dynamical rather than semantic: tokens do not encode meanings but are hypothesized to bias the local evolution of Φ through repeated interaction. Gradient

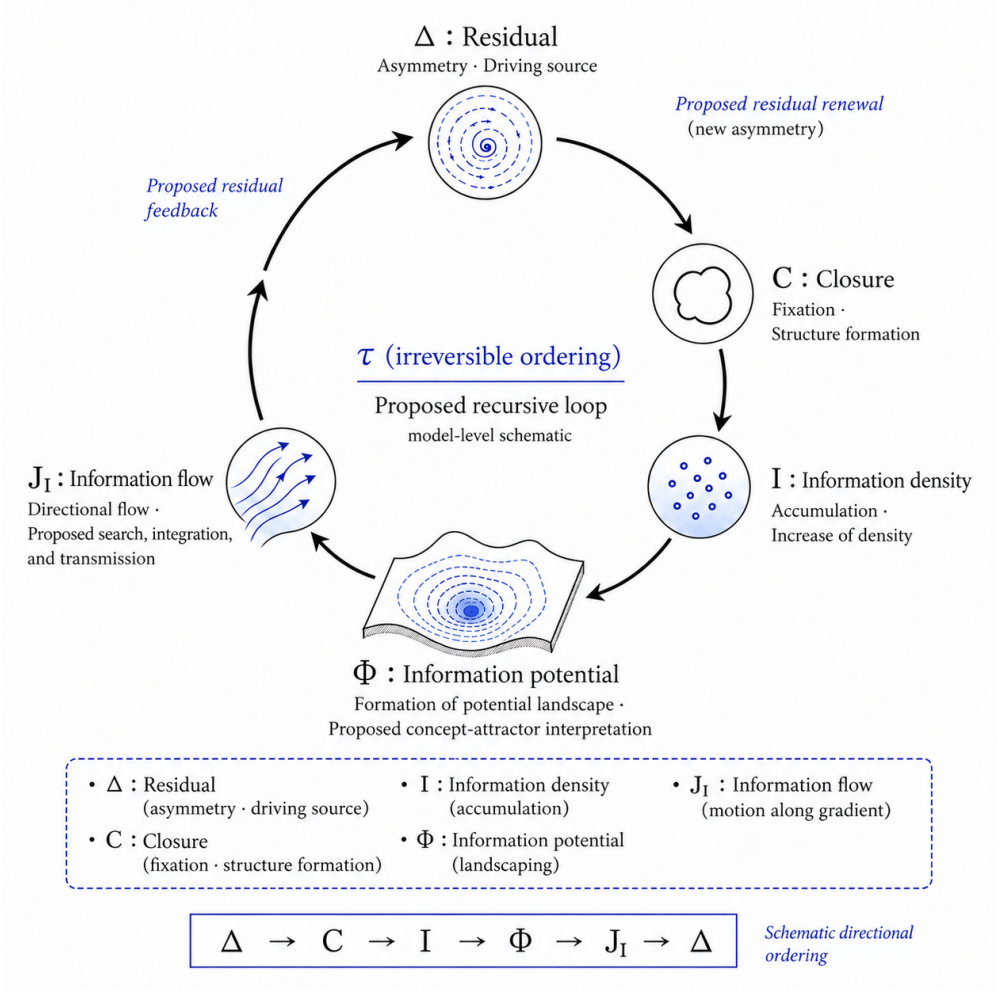


Figure 3: Proposed recursive loop in the VED×IFGT vocabulary. The schematic sequence $\Delta \to C \to I \to \Phi \to J_I \to \Delta$ links residual asymmetry Δ, closure C, information density I, potential structure Φ, and information flow J_I within the proposed feedback architecture. The ordering parameter τ expresses the model's irreversible-ordering premise; arrows are not a proof of causality or an executed dynamical witness.

reinforcement and well deepening are qualitative mechanism hypotheses, not consequences established merely by writing the term. The coupling expression remains schematic with respect to the numerical or vector realization of discrete tokens and the pairing implied by the dot notation: neither is formally specified. The mathematical kind of the effective, field, and token Lagrangian terms, and the coupling-parameter specification, also remain unresolved. No Euler–Lagrange or other variational or update dynamics are derived from this expression here.

4.3 Sedimentation and Conceptual Landscape Formation

The long-time expression below is retained as a sedimentation template for repeated token–field interaction, not as an established convergence result. Figure 3 summarizes the proposed feedback architecture with the framework's irreversible-ordering premise.

$$\Phi(x,\tau) \xrightarrow{\tau\to\infty} \Phi^*(x). \tag{9}$$

The arrow is a long-time template with no specified convergence mode. The following history interpretation retains unresolved formal and source-version status: how the displayed limit target relates to continued reshaping under renewed input is not decided here. The

nonuniqueness and nonstatic wording below is version-sensitive; no interpretation is selected to settle that difference.

Φ^* is the sedimented conceptual landscape. It is not necessarily unique or static; it reflects the history of all prior interactions and continues to be reshaped by new input, though at decreasing rate as the landscape matures.

4.4 Concept Definition

Definition 4.1 (Concept)**.** *A* concept *at position x is a locally stable region of the information potential satisfying:*

$$\nabla\Phi(x) \approx 0, \qquad \textit{with stability under small perturbations of } J_I. \tag{10}$$

The depth of the potential well at x is proposed as a structural correspondence for resistance to deformation; width is compared with contextual range. These interpretations are not semantic calibrations.

Definition 4.2 (Understanding)**.** Understanding *a conceptual region $\mathcal{R}$ is the state in which the traversal cost of J_I through $\mathcal{R}$ has been reduced to below a system-dependent threshold θ:*

$$\int_{\mathcal{R}} |\nabla\Phi|\, dx < \theta. \tag{11}$$

Here θ is a system-dependent effective threshold expressing the allowable range of recursive cost. Its concrete value depends on the structure of the system, including D, μ, α, and related parameters, and is not specified in the present work. Understanding is not a binary state but a dynamical condition that can be entered and exited as Φ evolves.

The definition specifies a local model state, not an independent cognitive or phenomenal criterion. Its proposed relation to felt ease must therefore be assessed separately; defining a low-cost state does not explain the experience of understanding.

4.5 The Strict Positivity Constraint

The strict-positivity statement is retained as a model-level admissibility condition. This positive-range constraint, the positive-output premise below, and preservation under the full balance evolution are separate formal burdens; their justification is not established here. The statement,

$$\boxed{I(x,\tau) > 0 \qquad \forall\, x,\, \tau,} \tag{12}$$

is stated with $C < C_{\max}$ and the positive-output premise $I = f(1-C) > 0$ for $C < C_{\max}$. Here $C_{\max}$ denotes the ideal complete-closure boundary, not an ordinary attainable state of the conceptual system.

Proposition 4.1 (No ordinary complete conceptual closure)**.** *Within the VED×IFGT model interpretation, $I > 0$ excludes complete conceptual closure as an ordinary dynamical state. Separately, in nontrivial landscapes with residual asymmetry, spatial inhomogeneity, or boundary structure, further gradients in Φ and available directions for J_I are proposed to reappear. This is a conditional differentiation hypothesis, not a consequence of positivity alone. The closure interpretation is not the pointwise mathematical claim that every positive I must have $\nabla\Phi \neq 0$. Without additional conditions, positivity alone does not imply nonzero flow either; the full flow includes more than the potential-gradient contribution.*

Remark 4.1. *This model-level distinction motivates a comparison with intelligence and understanding, rather than deriving them from positivity. Stable attractors may appear locally quiescent. Further differentiation is proposed to remain possible under new perturbations or wider observation windows; no unconditional learning or innovation guarantee is inferred.*

5 Structural Interpretation

5.1 Candidate Structural Classes

The VED×IFGT framework motivates the following candidate structural classes and cognitive interpretations. The table organizes a model-based classification, not a formal existence or classification theorem; detailed solution analysis and independent cognitive correspondences remain future work.

Candidate class	**Field structure**	**Cognitive interpretation**
Smooth gradient region	$\nabla\Phi$ slowly varying	Background cognitive disposition
Shallow metastable well	$\nabla\Phi \approx 0$, small depth	Tentative or unstable concept
Deep stable well	$\nabla\Phi \approx 0$, large depth	Entrenched concept or belief
Limit-cycle trajectory	J_I circulates without sedimentation	Recurring thought pattern
Barrier region	$\lvert\nabla\Phi\rvert$ large	Cognitive block or resistance
Ridge structure	Local maximum of Φ	Conceptual boundary or category edge

5.2 Depth, Width, and Robustness

Three geometric parameters characterize a conceptual attractor:

Depth ($-\min\Phi$ within the well) is proposed to correspond to resistance to deformation by new input: deeper wells are compared with more resistant concepts, shallower wells with more easily displaced ones.

Width (spatial extent of the sub-threshold region) is compared with contextual range: broad wells with wider applicability, narrow wells with more context-specific concepts.

Robustness (curvature of Φ at the minimum) is proposed as a comparative indicator of precision: higher curvature is associated qualitatively with sharper concepts, lower curvature with fuzzy or polysemous ones. None of these semantic attributes is calibrated by the geometric description alone.

These geometric attributes are not treated as fixed intrinsic properties; they are modeled as evolving with Φ under the relations in Eqs. (2)–(7).

5.3 Conceptual Boundaries

Ridge structures in Φ — regions of locally high potential separating neighboring wells — represent conceptual boundaries in this account: the distinctions between concepts. These boundaries are not sharp lines but extended regions of high traversal cost.

The qualitative hypothesis is that conceptual boundaries can shift under repeated linguistic or experiential perturbation. Tokens coupling near a ridge are proposed to reinforce it (deepening a distinction) or erode it (merging adjacent concepts). This is a candidate dynamical account of semantic drift and conceptual change, not an established cognitive mechanism or semantic mapping.

5.4 Vocabulary Selection and Distributional Description

The full set of stabilized tokens $\{\ell_i\}$ is called a vocabulary in this framework. The expression below selects a set by stability; the broader description of vocabulary as distributed across the field does not turn that set into a probability distribution:

$$\mathcal{V} = \{\ell_i \mid S(\ell_i) > \epsilon\}, \tag{13}$$

where $S(\ell_i)$ is a stability functional intended to depend on reuse frequency, persistence time, and basin depth, and ϵ is a system-dependent threshold. The functional and threshold are not given an extraction or calibration rule here.

Vocabulary is interpreted as a field-related organization reshaped by use. Token frequency is proposed to relate to basin accessibility, not to serve as a measure of semantic importance. Fuzzy, context-sensitive distinctions are part of this interpretive picture, not an operational token criterion.

6 Dynamics

6.1 Sedimentation

Sedimentation names the proposed process of well deepening and conceptual stabilization under repeated token–field interaction. The following three stages describe a functional feedback architecture, not an executed dynamical witness. The first token/basin bootstrap and the update consequences of coupling remain unresolved.

Stage 1: Initial perturbation. In the proposed scenario, a novel input Δ is associated with a σ_{drive} spike, a local increase in I, and a transient potential gradient. The information flow J_I is modeled as responding to the perturbation; this is not a reconstructed thought path.

Stage 2: Reinforcement. If the perturbation is repeated — through subsequent exposure to similar input or through internal recursive activation — the local gradient in Φ is hypothesized to be reinforced. Well deepening is a proposed effect of token–field coupling (8), not a result ensured by each reuse or by the expression alone.

Stage 3: Stabilization. The proposed stabilization stage associates well deepening with lower traversal cost and subsequent reuse with further reinforcement. A self-sustaining concept is the resulting maintenance hypothesis, not a demonstrated consequence of the feedback architecture.

6.2 Forgetting

Forgetting is interpreted here by analogy with relaxation of Φ toward a flatter configuration when reuse frequency drops. The existing dissipation term is retained as the formal template:

$$\sigma_{\text{dissip}} = \Gamma C. \tag{14}$$

In this model picture, reduced accessibility is associated with a redistribution of stability rather than erasure of stored symbols: shallowing wells are proposed to raise traversal cost. This does not establish that forgotten concepts never disappear. Reactivation (remembering) is correspondingly modeled as re-deepening through renewed token–field interaction, not identified with a universal memory mechanism.

6.3 Conceptual Change

Conceptual change — the shift in meaning, scope, or application of an established concept — is represented here using changes in Φ^*: the merger of two wells, the splitting of one well into

two, or displacement within the landscape. These geometric alternatives organize a proposed cognitive correspondence; they do not resolve the long-time target or prove a topology-to-cognition map.

The candidate account associates such changes with sustained perturbation relative to the depth of existing attractors, not with a calibrated necessary or sufficient condition for belief revision. The effective barrier term σ_{barrier} plays a protective role in this effective description: as $C \to C_{\max}$, the resistance interpretation is compared with increasing difficulty of changing a deeply entrenched concept. This should be understood as an effective description of overclosure within the conceptual observation window, not as an externally imposed wall. This offers a candidate account of resistance to belief revision; the reciprocal term does not establish cognitive prohibition.

6.4 Insight and the Rapid-Transition Hypothesis

The sudden sense that a problem has been resolved or disparate ideas connected motivates a proposed correspondence with rapid changes in the local topology of Φ. Merging wells or a collapsing barrier provide a bifurcation-like representation. A geometric transition, cognitive problem resolution, and a felt event are distinct claims; no bifurcation theorem or independent cognitive endpoint is supplied here.

The subjective sense of suddenness is hypothesized to correspond to the threshold picture of accumulating perturbations followed by a rapid transition. The picture does not independently identify perceptual thresholds or explain felt suddenness merely by naming a transition.

7 Scale Inversion: Individual to Social

7.1 Comparing Functions and Dominant Costs

The preceding model is interpreted at the level of an individual cognitive system. In that interpretation, recursion and compression are the proposed dominant costs: revisiting states and representing continuous configurations with reusable discrete handles. Language tokens are assigned an internal-control role under these architectural assumptions.

A social extension is proposed by comparing this role with synchronization and coordination among interacting agents. Tokens are hypothesized to help stabilize shared trajectories and reduce uncertainty about other agents' responses. The table contrasts the proposed dominant costs and token functions; it does not establish that individual and collective dynamics are identical:

Scale	**Dominant cost**	**Token function**
Individual	Recursion, compression	Internal control units
Social	Synchronization, coordination	Shared landscape anchors

The comparison proposes a similar re-addressing function under different effective constraints. It does not establish unchanged token properties across scales. Collective aggregation, a common state space, and a typed translation remain unspecified; no shared-field measurement follows from reusing the same notation.

7.2 Social Conceptual Landscape

The social hypothesis compares repeated exposure to shared linguistic inputs, given sufficiently similar prior landscapes, with correlated well deepening. Shared conceptual landscape is the model-level name for this proposed organization.

In this picture, shared terrain is distributed across agents and maintained through ongoing linguistic interaction, rather than stored in one agent. History-shaped collective geometry is an interpretation of those interactions, not a measured collective field.

7.3 Norms, Institutions, and Paradigms

Deeply sedimented social regions are compared structurally with norms, institutions, and scientific paradigms. This attractor analogy uses the collective Φ to represent persistence; it is not an identity between institutions and formal attractors.

A separate hypothesis relates depth to stability and deviation cost, and coordinated perturbation to institutional change. It offers a way to consider persistence without making central enforcement the sole explanation, not evidence that enforcement is unnecessary. Historical difficulty of paradigm shifts is a possible comparison target, not a result explained by well depth. Independent institutional criteria and depth estimation remain unresolved; stability cannot serve as both the definition of depth and independent evidence for its explanatory role.

7.4 Intelligence as Landscape Maintenance

Within this social-scale account, we use a local functional designation for the proposed distributed process:

> *Social intelligence, as used here, names distributed maintenance, synchronization, and reshaping of conceptual landscapes through language.*

This naming is not an established general definition or generation of intelligence, nor does it exclude individual intelligence. Its relation to the repository's Intelligence theories is a conceptual handoff, not validation transferred from downstream work.

Individual agents are modeled as transient participants in this process: they traverse the landscape, introduce local deformations, and transmit stabilized structures to subsequent agents through externalized linguistic logs.

The strict positivity $I > 0$ motivates an analogous nonclosure reading for the proposed social landscape, not a derivation of collective dynamics. Further differentiation and innovation are proposed possibilities under new perturbations or wider observation windows that reactivate residual asymmetries, not outcomes guaranteed by positivity alone.

8 Pathologies and Limits

8.1 Deliberate Exclusions

The present framework deliberately excludes the following:

Semantic theory. No claims are made about reference, truth conditions, or interpretative content. Meaning, where it appears, is an emergent label applied to stable trajectories; it is not a primitive of the theory.

Phenomenology and consciousness. The framework does not generate or explain subjective experience. Conscious access is treated as a downstream topic for comparison with Consciousness Theory, whose source is now present in the repository. That comparison neither transfers a mechanism into CT nor validates its phenomenal interpretations.

Neurobiological implementation. The theory describes structural and dynamical conditions at a level of abstraction that is independent of specific neural architectures. Correspondence to neural implementation is deferred to future work.

8.2 Proposed Failure Regimes

The following are qualitative regime hypotheses, not diagnostic or clinical classifiers, nor necessary or sufficient conditions for cognitive states.

Overclosure (too-deep wells). A large effective σ_{barrier} relative to σ_{drive} is interpreted as a regime resistant to landscape change. The proposed cognitive correspondence is more repetitive, less adaptable thought. Rigidity, dogmatism, and entrenched belief are phenomenological comparison terms here, not states identified or causally explained by the parameters.

Underclosure (too-shallow wells). Dominance of σ_{dissip} is associated in the model picture with shallowing faster than well formation. This is proposed to correspond to unstable thought and difficulty stabilizing concepts. A relation to less durable understanding is a qualitative hypothesis, not an established prevention mechanism.

Synchronization breakdown (social scale). Divergent prior landscapes, insufficient shared exposure, or deliberate manipulation are proposed to weaken synchronization across agents. Fragmentation of shared terrain is a model interpretation of this pattern, offering a candidate comparison with communication failure, cultural conflict, and epistemic polarization. These outcomes are not identified with one measured regime or inferred from collective geometry alone.

8.3 Scope Limits

The theory proposes structural and dynamical descriptions; it does not establish:

- Quantitative predictions about specific cognitive systems (this requires concrete specification of K, D, μ, α, Γ, κ_B).
- Correspondence conditions between Φ-geometry and observable neural or behavioral data.
- A complete theory of language acquisition, semantic change, or collective intelligence.

These remain as future work within the proposed framework.

9 Relation to the Note Series

The source of this paper refers to a companion note series (L1 layer) using sedimentation, flow, and terrain vocabulary. The exact note artifact, title, and version remain unresolved. The table preserves CT's own educational comparison between that recorded vocabulary and its model structure; it is not a verified reconstruction of an independently identified note, a typed map, or validating evidence. The cognitive and phenomenal entries are proposed interpretations, not identities established by the displayed correspondence.

Note vocabulary recorded here (J: Japanese)	Proposed model correspondence
log no chinden^ "log sedimentation"	Deepening of Φ via σ_{drive}
anteiten "stable point"	Locally stable region: $\nabla\Phi \approx 0$
gainen no kubomi "conceptual hollow"	Potential well in Φ
zentai koubai "overall gradient"	Information potential $\Phi = K * I$
tooriyasusa "passability"	$-\Phi(x, \tau)$ (high passability = low Φ)
tani "valley"	Deep well in Φ
kabe "wall"	Ridge region of high Φ
rikai shita kankaku "sense of understanding"	Reduction of traversal cost $\int \lvert\nabla\Phi\rvert dx$
shikou no kuse "habitual thought pattern"	Limit-cycle trajectory of J_I
GPU-teki na souchi "GPU-like device"	Sub-threshold operation of Φ on J_I
gengo ryuushi "language particle"	Discrete token ℓ_i coupling to $\nabla\Phi$
asshuku "compression"	Quantization operator $\mathcal{Q}$
shakai-teki kyouyu chikei "shared social terrain"	Proposed social Φ^* maintenance across agents

9.1 Limits of This Comparison

The descriptions recorded here are used as nontechnical intuitions, without attributing verified commitments to an unidentified note artifact. From the standpoint of CT they are *interpretations* of field-theoretic quantities rather than definitions.

In particular:

- The recorded meaning vocabulary is compared with repeated traversal and potential-well depth, not with an intrinsic semantic object or a calibrated semantic measure.
- The recorded sense-of-understanding description is proposed to correspond to changes in ease associated with (11). The local definition is not independent evidence for this phenomenal correspondence.
- The CPU/GPU distinction attributed here to a Consciousness note is compared with a fast-lane / slow-lane decomposition. Consciousness Theory now has repository source, but the exact note-to-paper version relation remains unresolved. This is a structural comparison, not a shared quantity, typed translation, or transfer of a demonstrated mechanism.

9.2 The L3 → L1 Direction

The present paper (L3) does not revise or replace the note series (L1). Its contribution here is interpretive organization: a model vocabulary for comparing the recorded intuitions, not a theoretical validation of them. The exact source attribution remains open.

Terrain, flow, and sedimentation are compared with field-theoretic objects. Understanding, meaning, and thought are separate interpretive readings of that vocabulary, not independently explained by assigning formal referents. The two exposition layers can be complementary without either providing validating evidence for the other.

10 Discussion

10.1 Relation to Existing Frameworks

Dynamical systems approaches to cognition. The treatment of concepts as attractors in a dynamical system has precedent in connectionist and embodied cognition research [e.g., Thelen and Smith, 1994]. The present framework draws structural motivation from VED and the effective dynamical conventions of IFGT, and proposes an explicit token–field coupling expression. This source reuse does not derive cognitive attractors from an upstream axiom or establish their mechanism.

Distributional semantics. The comparison with distributional semantics is conceptual: token-space geometry associated with co-occurrence is contrasted here with CT's proposed history-dependent token–field geometry. CT emphasizes *use*, without claiming that use and *co-occurrence* are empirically exclusive or that this comparison establishes a superior semantic representation. No corpus, embedding metric, or cross-model mapping is supplied here.

Predictive processing and free energy. Predictive processing accounts treat cognition as the minimization of prediction error [Friston, 2010]. A conceptual comparison is proposed: σ_{drive} may be interpreted alongside prediction error influx, and deepening of Φ wells can be understood by analogy with long-term error minimization. CT does not identify these quantities with prediction-error or free-energy variables. It does not commit to a Bayesian generative model, and this noncommitment is not a proof of greater generality or theoretical inclusion.

10.2 Open Questions

Quantitative correspondence. The framework proposes structural descriptions but does not specify the concrete form of the convolution kernel K or the parameter values D, μ, α, Γ, κ_B. Identifying these from empirical data — behavioral, linguistic, or neural — is a primary open question.

Topology and concept individuation. The framework defines concepts as locally stable regions of Φ, but the conditions under which a region counts as a single concept rather than a cluster of related concepts are not yet specified. A topological criterion for concept individuation is needed.

Multi-agent synchronization dynamics. The social landscape is described qualitatively; a formal treatment of multi-agent token–field interaction, including conditions for stable shared landscape formation and breakdown, remains to be developed.

Connection to neural implementation. The correspondence between the field quantities I, Φ, J_I and measurable neural quantities (firing rates, connectivity weights, oscillatory dynamics) is a major open question that requires collaboration with empirical neuroscience.

11 Conclusion

This paper has developed Conceptual Topography (CT), a field-theoretic account of conceptual landscapes, as the first application layer of the VED×IFGT framework.

The central proposal uses history-shaped geometry of the information field $I(x, \tau)$ under recursive token–field interaction as a model-local conceptual landscape. This designation is separate from convergence, a cognitive mechanism, and a claim about access to experience. No semantic primitives, intentional states, or meaning-bearing units are required within this model.

The retained definitions and proposed interpretations are:

1. A concept is a locally stable region of the information potential $\Phi = K * I$, defined by $\nabla\Phi \approx 0$ with stability under perturbation. Its depth, width, and robustness are dynamical quantities that evolve continuously with the history of interaction.

2. Understanding is a reduction in the traversal cost of a conceptual region, not an epistemic achievement. It can be entered and exited as the landscape evolves.

3. Language tokens function as discrete re-addressable units that couple to $\nabla\Phi$ without carrying semantic content. Low-cost recursion motivates this architectural role; deepening through repeated interaction remains a qualitative hypothesis, not a guaranteed consequence of coupling.

4. Within the model, strict positivity $I(x, \tau) > 0$ is interpreted as excluding ordinary complete conceptual closure. Local quiescence is allowed. Further differentiation under new perturbations or wider observation windows is a conditional proposal, not a pointwise consequence of positivity.

5. At the social scale, comparable token functions under different costs motivate a shared-landscape hypothesis. Norms, institutions, and paradigms are compared with attractors; depth/stability relations remain hypotheses requiring independent social criteria. Collective maintenance is local functional vocabulary, not an established general definition of intelligence.

The framework does not revise or replace the companion note series. It preserves CT-local educational comparisons with recorded intuitions, not validating support for an independently identified note. The exact note and Master-table source/version attributions remain unresolved.

Detailed developments — quantitative parameter identification, topological criteria for concept individuation, multi-agent synchronization dynamics, and neural correspondence — remain as future work. The long-time target, coupling signature, positivity preservation, and other formal OPEN questions have not been resolved by scoping the prose. Likewise, separating understanding from felt ease and insight from subjective suddenness does not establish either correspondence.

In sum, Conceptual Topography proposes field-theoretic representations of conceptual structure and cognitive functions, drawing on the VED×IFGT framework. Concepts and physical particles may be compared as projected stabilization regimes, not as identical entities or reducible objects. A recurring generative grammar is a heuristic motif in this comparison, not a shared-equation theorem. Depth, width, persistence, and observation window organize the comparison without establishing common quantities, a common field, or a typed translation. No empirical validation or explanation of subjective experience is claimed.

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